

Unit 34: Indeterminate Forms and L'Hôpital's Rule

1. The idea, in plain words

Back in Unit 2, you learned the “ $\frac{0}{0}$ fix-it kit” — factor and cancel, or rationalize, to dodge a dead-end limit. This unit hands you a much more powerful tool that works even when factoring is impossible: L'Hôpital's Rule.

Here's the surprising fact at the heart of it: if a limit of the form $\frac{f(x)}{g(x)}$ comes out to $\frac{0}{0}$ (or $\frac{\infty}{\infty}$) when you plug in the target value, you can replace the numerator and denominator with their own separate derivatives, and the limit of that new fraction will be exactly the same as the original limit. Keep doing this — differentiate top and bottom separately — until the indeterminate form finally clears up into something you can just plug straight into.

Picture why this makes sense: near the point in question, both the top and bottom of the fraction are shrinking toward zero (or blowing up toward infinity) together. What actually matters for the limit isn't their sizes — it's how fast each one is changing right at that instant. That's precisely what a derivative measures. So comparing their derivatives is like comparing their “closing speeds” instead of comparing two numbers that are both stuck at zero.

A critical warning before you dive in: L'Hôpital's Rule only applies when the limit is genuinely indeterminate — specifically $\frac{0}{0}$ or $\frac{\infty}{\infty}$. If you plug in the value and get an ordinary number (even something like $\frac{3}{1}$), stop — just report that number as the answer. Applying the rule to a non-indeterminate limit doesn't just waste effort, it can actually produce the wrong answer.

Another crucial warning: L'Hôpital's Rule is NOT the quotient rule. You are differentiating the numerator and denominator completely separately — you do not combine them into one single derivative using the quotient rule formula. That would be solving a totally different problem.

2. Toolbox

L'Hôpital's Rule: if $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ produces the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and f, g are differentiable near a , then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

(provided the limit on the right exists, or is $\pm\infty$).

The procedure:

1. Plug in the target value. Confirm you genuinely get $\frac{0}{0}$ or $\frac{\infty}{\infty}$ — if not, stop and just report the direct answer.
2. Differentiate the numerator and denominator separately (not with the quotient rule).
3. Take the limit of this new fraction.
4. If it's still indeterminate, repeat the process — apply L'Hôpital again on the new fraction.
5. If it's no longer indeterminate, plug in and report the answer.

Recognizing $\frac{\infty}{\infty}$ forms: these work exactly the same way as $\frac{0}{0}$ forms — differentiate top and bottom separately, and check the new limit.

3. Common mistakes

- Applying L'Hôpital's Rule to a limit that isn't actually indeterminate. Always check first — if plugging in gives you a normal finite number over a nonzero number, that number is the answer; don't touch the derivative.
- Confusing L'Hôpital's Rule with the quotient rule. You differentiate the top and bottom independently, each on its own — you never combine them into a single derivative expression.
- Stopping after one application when the result is still indeterminate. Always re-check the new fraction — sometimes you need to apply the rule two or even three times before it resolves.
- Forgetting basic limits you already know how to compute directly (like recognizing $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$) and reaching for L'Hôpital unnecessarily when a quicker method is available — not wrong, just slower.
- Losing track of algebra when a limit is a product of two things both going to 0 (a “ $0 \times \infty$ ” style setup in disguise). Sometimes it's easier to split the limit into two separate, recognizable pieces (using limit laws) rather than brute-forcing repeated derivatives on a tangled product.

4. Problem Set

Warm-up

1. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$
2. $\lim_{x \rightarrow 0} \frac{\sin x}{x}$
3. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$
4. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$
5. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$
6. $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$

Standard

7. $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 1}$
8. $\lim_{x \rightarrow 0} \frac{x^2 + 3}{x + 1}$ (check carefully whether this is actually indeterminate before doing anything!)
9. $\lim_{x \rightarrow 0} \frac{\tan x}{x}$
10. $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$
11. $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{3x}$
12. $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$

Challenge

13. $\lim_{x \rightarrow 1} \frac{x^3 - 1}{4x^3 - x - 3}$

14. $\lim_{t \rightarrow 0} \frac{\sin(t^2)}{t^2}$
15. $\lim_{x \rightarrow 1} \frac{x - 1}{\ln x - \sin(\pi x) + x - 1}$
16. $\lim_{x \rightarrow 0} \frac{\tan^2 x (1 - \cos x)}{x^4}$
17. $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

Applied

18. Show, using L'Hôpital's Rule, that $\lim_{t \rightarrow a} \frac{t^2 - a^2}{t - a} = 2a$, and explain how this confirms the derivative of $f(t) = t^2$ at $t = a$.
19. For $C(x) = x^2 + 3x$, evaluate $\lim_{x \rightarrow 0} \frac{C(x) - C(0)}{x}$, and confirm it matches $C'(0)$.
20. Show, using L'Hôpital's Rule, that $\lim_{t \rightarrow 0} \frac{1 - e^{-kt}}{t} = k$.
21. Find $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(5x)}$.